

A SHORT BOUND FOR DIRECTED LOCAL CONNECTIVITY

ABSTRACT. A theorem of Huang and Lyu on repeated directed two-paths immediately improves the known error term for digraphs of bounded local connectivity. For every fixed $m \geq 3$, both the arc- and vertex-disjoint extremal numbers are at most $n^2/4 + (m-1)n + O_m(1)$.

Let $P_{s,2}$ denote the union of s directed paths of length two with common initial and terminal vertices. Huang and Lyu proved the following sharp estimate for sufficiently large order.

Theorem 1 (Huang–Lyu [1]). *For $t \geq 2$ and*

$$n \geq N(t) := \max \left\{ t^3 + 4t^2 + 3t + 4, \frac{17}{2}t^2 + 30t + 27 \right\},$$

every $P_{t+1,2}$ -free simple digraph has at most

$$(1) \quad G(n, t) := \left\lceil \frac{n+t}{2} \right\rceil \left\lfloor \frac{n-t}{2} \right\rfloor + tn + 1$$

arcs.

For a simple digraph D , let $\lambda_D(u, v)$ be the maximum number of arc-disjoint directed u – v paths and let $\kappa_D(u, v)$ be the maximum number of internally vertex-disjoint such paths. Write $\ell_m^{\text{dir}}(n)$ and $k_m^{\text{dir}}(n)$ for the corresponding extremal arc counts under the ceiling $m-1$.

Corollary 2. *Let $m \geq 3$ and $n \geq N(m-1)$. Then*

$$\begin{aligned} \ell_m^{\text{dir}}(n) &\leq k_m^{\text{dir}}(n) \leq G(n, m-1) \\ &= \left\lceil \frac{n+m-1}{2} \right\rceil \left\lfloor \frac{n-m+1}{2} \right\rfloor + (m-1)n + 1. \end{aligned}$$

Consequently, for each fixed $m \geq 3$,

$$\ell_m^{\text{dir}}(n), k_m^{\text{dir}}(n) \leq \frac{n^2}{4} + (m-1)n + O_m(1).$$

Proof. If D contained m directed two-paths $u \rightarrow x_i \rightarrow v$ with common endpoints, then their distinct internal vertices would make the paths both arc-disjoint and internally vertex-disjoint. Thus either connectivity ceiling makes D $P_{m,2}$ -free, and Theorem 1 applies with $t = m-1$. Finally $\kappa_D(u, v) \leq \lambda_D(u, v)$ gives $\ell_m^{\text{dir}}(n) \leq k_m^{\text{dir}}(n)$, while

$$\left\lceil \frac{n+t}{2} \right\rceil \left\lfloor \frac{n-t}{2} \right\rfloor \leq \frac{n^2 - t^2}{4}$$

gives the asymptotic form. □

Remark 3. The augmented bipartite construction gives

$$\ell_m^{\text{dir}}(n) \geq \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor = \frac{n^2}{4} + \frac{m-2}{2}n + O_m(1).$$

Hence Corollary 2 reduces the previous upper coefficient $4(m-1)$ to $m-1$, but it does not determine the conjectured coefficient $(m-2)/2$: the two-path reduction deliberately discards all information about direct arcs and longer routes.

REFERENCES

- [1] Z. Huang and Z. Lyu, *Extremal digraphs containing at most t paths of length 2 with the same endpoints*, Discrete Appl. Math. **388** (2026), 1–10.